Jeopardy! Wagering Strategy

A wagering-strategy model and practice tool for Jeopardy!'s two decisions that carry real stakes: the **Daily Double** and **Final Jeopardy**. Built for 3-player games, with 2-player and 1-player (solo practice) as reduced cases of the same model.

Not affiliated with Jeopardy! or Sony Pictures Television. A study and post-game analysis tool only — not for use during an actual taping.

What's here

Path	What it is
`app/index.html`	Interactive browser trainer: calculators + a graded quiz mode. Live at https://willhoop.github.io/jeopardy-wagering/app/, or open the file directly in any browser, no server needed.
`engine/`	The Python engine the trainer's logic is based on: `fj_wager.py` (Tier 1, guarantee-based), `fj_bestresponse.py` (Tier 2, confidence-weighted), `simulate.py` + `dd_wager.py` (Daily Double, Monte Carlo).
`engine/behavior.py`	Simulated contestant behavior and the tie rule, in one configuration block: role-based Final Jeopardy wager mixes, accuracy by score rank, correlated right/wrong outcomes, and Daily Double behavior. Each value is marked CALIBRATED, PUBLISHED or ASSUMED.
`tests/`	46 Python tests, plus `js_parity.js`, which checks the app's JavaScript engine against the same hand-derived values. Hand-derived expected values where possible; invariant checks where Monte Carlo output can't be hand-derived. CI (`.github/workflows/ci.yml`) runs both on every push and pull request.
`docs/white-paper.md`	The full research, math, and methodology writeup. Start here to understand *why*, not just *what*.
`docs/thesis/underwagering.md`	A separate thesis: why strong players underwager, what it costs, and when betting less is right.
`docs/deck.pptx`	A plain-English overview, no math required. Links `white-paper.md` on the last slide.
`docs/tutorial/`, `docs/how-to/`, `docs/reference/`, `docs/explanation/`	Task-oriented documentation (Diátaxis structure), written in ASD-STE100.
`docs/adr/`	Architecture decision records — the reasoning behind data-sourcing and opponent-model choices.
`docs/adr/0003-simulated-contestant-behavior.md`	The opponent model structure and tie rule: sources, decision, and negative consequences. Supersedes ADR-0002.

Path	What it is
`docs/adr/0004-modern-era-calibration.md`	The calibration of the simulated opponents to modern-era play (Seasons 38-41, 2021-2025), and why no source statistic is published. Supersedes ADR-0003's accuracy, correlation, leader-cover and Daily Double values.
`RISK_REGISTER.md`	Open risks and known limitations, tracked live.
`CHANGELOG.md`	Every change, in Keep a Changelog format.
`build/`	Generated PDF copies of every doc, and `md_to_pdf.py`, the script that builds them. Markdown is the source of truth; regenerate the PDF if the source changes.
`llms.txt`	A short summary and links for AI agents (llmstxt.org format).

Quick start

Open the trainer at https://willhoop.github.io/jeopardy-wagering/app/, or open `app/index.html` locally.

Run the tests:

```
cd tests
python -m pytest
cd ..
node tests/js_parity.js
```

Read the theory first if you want to understand the math before using the tool: `docs/white-paper.md` (or its PDF twin in `build/`).

Two ways every wager is calculated

1. **Safe Bet (Tier 1)** — the wager that can't hurt you if you're right, assuming every opponent plays the worst realistic case against you. No confidence estimate needed.
2. **Smart Bet (Tier 2 / Daily Double engine)** — factors in your actual confidence and plays you against simulated contestants to maximize your real win chance. A tie for first goes to a tiebreaker clue (the rule since November 2014), so it counts as half a win. The simulated rivals are calibrated to modern-era play (Seasons 38-41, 2021-2025): they hunt Daily Doubles and bet them hard, the leader almost always covers, and players in front answer Final Jeopardy correctly more often. Second place's wager mix still comes from a published study of pre-2013 games (Tesauro et al., 2013); the rest are labeled assumptions — see `docs/adr/0003-simulated-contestant-behavior.md` and `docs/adr/0004-modern-era-calibration.md`. The calibration source (J! Archive) may not be republished, so no calibrated value appears in the documents.

Status

Engine and tests: validated by hand-derived cases, with CI on every push. Tier 2 opponent model: partly calibrated to modern-era play, partly published (older games), partly assumed. See `RISK_REGISTER.md` for what is proven versus assumed, and `CHANGELOG.md` for what shipped when.

License

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